On efficacy of Meta-Learning for Domain Generalization in Speech Emotion Recognition

Raeshak King Gandhi · **Vasileios Tsouvalas** · Nirvana Meratnia

Eindhoven University of Technology

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01 Context & approach

The challenge

Speech emotion models often fail on unfamiliar speakers, languages and corpora. *DA-MLDG* tests whether weighting learning by estimated shifts between training domains improves recognition on a completely held-out corpus.

Our contribution

Estimate domain shift from the gap between average energy scores.
Use that gap to weight the meta-test loss during training.
Compare seven transfer scenarios and expose remaining generalization failures.

1 Split the source corpora

Sample disjoint meta-train and meta-test domains; keep the final target corpus outside training.

2 Adapt and measure shift

Take a meta-train gradient step, then measure the absolute energy-score gap between the two source subsets.

3 Optimize for transfer

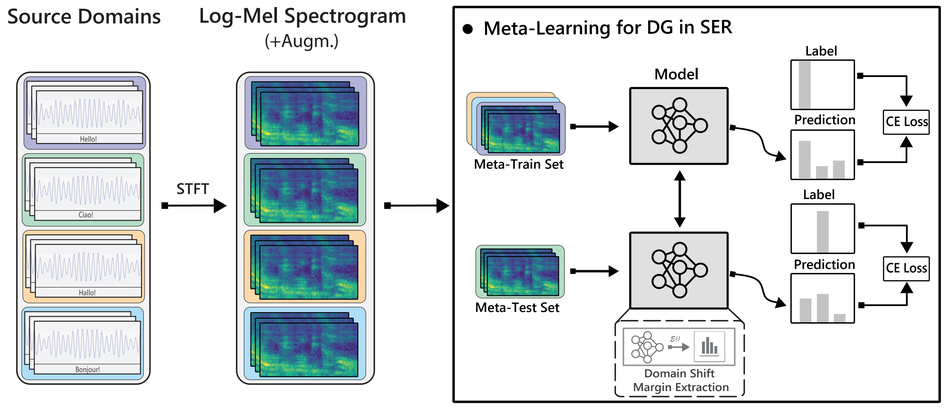
Combine meta-train loss with the gap-weighted meta-test loss; update the model and resample the source split.

Scope

Transfer gains are uneven; the reported performance remains insufficient for practical use. Small corpora, missing classes and no trial dispersion limit comparisons.

02 Method

Domain-aware meta-learning exposes persistent limits in cross-corpus speech emotion recognition.



Core formulation

$$\mathcal{L} = \mathcal{L}_{\theta}(\mathcal{S}') + \gamma \mathcal{L}_{\theta'}(\mathcal{V})$$

S′ and V are disjoint source subsets. θ′ follows a meta-train update; γ controls the weight of its meta-test loss.

$$\gamma = \left| \mathcal{E}(\mathcal{S}', p_{\theta'}) - \mathcal{E}(\mathcal{V}, p_{\theta'}) \right|$$

E is the absolute mean log-sum-exp of temperature-scaled logits. A larger source-domain gap gives the meta-test loss more weight.

Paper

Code

03 Results

4 / 7
above supervised Deep-All
Log-mel transfer scenarios

2 / 7
highest among four methods
Log-mel scenarios 4 and 5

Cross-domain emotion recognition

Scenario / target	Deep-All	RSC	MLDG	DA-MLDG (Ours)
1 / IEMOCAP	26.95	32.36	27.61	**31.35**
2 / RAVDESS	43.40	44.78	38.86	**39.41**
3 / TESS	47.03	43.77	46.29	**46.80**
5 / AESDD	60.65	56.69	60.45	**62.26**
6 / ShEMO	31.54	40.70	37.87	**38.79**

Held-out target accuracy (%), mean of ten trials; log-mel inputs with a VoxCeleb-pretrained ResNet-18.

Transfer with pretrained audio embeddings

Scenario / target	Deep-All (%)	DA-MLDG (%) (Ours)
1 / IEMOCAP	28.57	**32.11**
2 / RAVDESS	42.61	**39.82**
5 / AESDD	61.17	**64.11**
6 / ShEMO	33.16	**40.22**

The same source/target scenarios using TRILLsson embeddings; mean target accuracy (%) over ten trials.

Takeaway

Transfer improves unevenly: *DA-MLDG* beats supervised Deep-All in four of seven log-mel scenarios and leads all four methods in two. With pretrained embeddings, it beats the supervised baseline in **only three of seven** scenarios.